

## Topic 1.8 - Rational Functions and Zeros

### Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

#### Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

#### Learning focus

- Determine rational-function zeros while preserving original domain restrictions.
- Use multiplicity and sign charts to describe behavior near zeros.
- Construct rational functions from specified zeros, holes, asymptotes, and scale conditions.

### Section A: Foundation and representation

#### Question 1

Foundation

No calculator

Find zeros and exclusions of  $\frac{(x-5)(x+2)}{(x+2)(x-1)}$ .

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#### Question 2

Factoring

No calculator

Find zeros, exclusions, and multiplicities of  $\frac{(x+1)^3(x-4)^2}{(x-2)(x^2+1)}$ .

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## Section A continued

## Question 3

Classification

No calculator

For  $R = \frac{x^2-4}{x^2-5x+6}$ , distinguish zeros, holes, and vertical asymptotes.

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## Question 4

Multiplicity

No calculator

Determine whether the graph crosses or touches at each zero of  $\frac{(x-1)^4(x+3)}{(x+5)^2}$ .

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## Question 5

Sign analysis

No calculator

Create a sign chart for  $\frac{(x+2)(x-1)}{x-4}$ .

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**Section B: Application and reasoning****Question 6**

Parameter

No calculator

Find  $k$  if  $R = k \frac{(x-2)(x+3)^2}{(x-1)(x+4)}$  and  $R(0) = 9$ .

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**Question 7**

Construction

No calculator

Construct a rational function with only one real zero 3, of multiplicity 2, and one vertical asymptote  $-1$ .

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**Question 8**

Error analysis

No calculator

A student cancels  $x - 2$  and then calls  $x = 2$  a zero of the simplified numerator. Explain the correct classification.

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## Section C: Challenge and error analysis

## Question 9

Reasoning

No calculator

A rational function has zero 1 of multiplicity 3 and no vertical asymptote at 1. What sign behavior occurs there?

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## Question 10

Challenge

No calculator

Construct a rational function with zeros  $-2$  double and  $4$  simple, hole at  $0$ , and vertical asymptote  $3$ .

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## Question 11

Context

No calculator

For  $P(t) = \frac{(t-2)(t-8)}{(t+1)(t^2+4)}$  on  $0 \leq t \leq 10$ , identify zeros and sign intervals.

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## Question 12

Logic challenge

No calculator

Is it possible for a degree-3 numerator with real coefficients to have exactly two distinct nonreal zeros and no real zero? Justify.

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## Section D: AP-style multiple choice

## MCQ 1

No calculator

What is the zero of  $\frac{(x-3)(x+5)}{(x-3)(x-1)}$ ?

- A.  $-5$  only
- B.  $3$  only
- C.  $-5$  and  $3$
- D.  $1$  only

## MCQ 2

No calculator

At a numerator factor  $(x + 2)^4$ , with denominator nonzero at  $-2$ , the graph

- A. crosses
- B. touches without sign change
- C. has a hole
- D. has a vertical asymptote

## MCQ 3

No calculator

Which factor can add degree without adding real zeros?

- A.  $x - 1$
- B.  $x^2 - 1$
- C.  $x^2 + 1$
- D.  $x^2 - 4x + 3$

## MCQ 4

No calculator

A canceled factor  $(x - a)$  means  $x = a$  is

- A. automatically a zero
- B. still excluded
- C. always a vertical asymptote
- D. in the domain after cancellation

**AP-style free response 1****No calculator - 6 points**

Let  $F(x) = \frac{(x-4)^2(x+1)}{(x-2)(x+1)(x^2+1)}$ .

**A.** State domain exclusions and simplify.

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**B.** Identify zeros and multiplicities.

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**C.** Identify holes and vertical asymptotes.

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**D.** Determine sign intervals and behavior at the zero.

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**AP-style free response 2****Calculator permitted - 6 points**

On  $0 \leq t \leq 10$ ,  $P(t) = \frac{(t-2)(t-8)}{(t+1)(t^2+4)}$  models a signed performance index.

**A.** Find all zeros and interpret them as threshold times.

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**B.** Determine intervals where the index is positive and negative.

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**C.** Calculate  $P(5)$  to three decimals.

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**D.** Explain why the denominator creates no domain problem on the contextual interval.

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